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A District-Wide Field Study in High School Government Classes

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Abstract

In a study conducted across an urban school district, we tested a classroom-based intervention in which students were taught online evaluation strategies drawn from research with professional fact checkers. Students practiced the heuristic of *lateral reading:* leaving an unfamiliar website to search the open Web before investing attention in the site at hand. Professional development was provided to high school teachers who then implemented six 50-minute lessons in a district-mandated government course. Using a matched control design, students in treatment classrooms (n = 271) were compared to peers (n = 228) in regular classrooms. A multilevel linear mixed model showed that students in experimental classrooms grew significantly in their ability to judge the credibility of digital content. These findings inform efforts to prepare young people to make wise decisions about the information that darts across their screens.

Keywords: Media literacy, digital literacy, civic education

Educational Impact and Implications Statement

This study tested the effectiveness of an intervention that taught high school students to make sound decisions on the Internet. Less than six hours of classroom instruction significantly improved students' judgment about the credibility of online sources.

Modern citizens scrolling through social media face an endless stream of appeals. Organizations solicit donations, volunteers, and signatures on petitions. Arguments in comment sections cry out for responses. There are news articles to pass on, Instagram posts to like, tweets to re-tweet. There aren't enough hours to research every issue. Facing a digital onslaught, people need efficient, time-saving strategies to separate good information from bad. What ways of online reading increase the chances that brief searches lead to informed decisions, rather than ones that leave users dazed and confused?

To be sure, there are times when what we search for requires a more protracted investigation: a medical problem that threatens the health of a loved one or an issue that must be defended in one's PhD orals. But many searches, the brisk forays that help us judge the messages that dart across our screens, tend to be variations of "not very long"—in some cases, minutes; in others, even less (Liu et al., 2010; Nielsen, 2011). What happens when our quick research goes awry? Repercussions may be slight when someone circulates a misleading post to a circle of friends. But in aggregate, false information shared thousands of times pollutes the information stream and diminishes trust in social institutions (Diamond, 2019; Kavanagh & Rich, 2018).

The present study tested an intervention aimed at improving students' ability to make quick but accurate judgments of Internet sources. We call our curricular approach *Civic Online Reasoning* (Wineburg et al., 2016), a term that distinguishes it from the broader fields of media and digital literacy, which embrace everything from creating videos to understanding the implications of one's digital footprint. Civic Online Reasoning narrows the focus by providing tools to efficiently sort through the avalanche of social and political messages that stream across our screens. Drawing on our prior research with professional fact checkers (Wineburg & McGrew, 2016, 2017, 2019), as well as interventions conducted at the high school and college level (Breakstone, Smith, Connors, et al., 2021; McGrew, 2020; McGrew et al., 2019), we designed a curriculum module of six 50-minute lessons and provided high school teachers with professional development to implement it. We then tested whether this module, delivered by regular teachers, could improve students' ability to accurately judge the credibility of online sources.

Heuristics of Online Evaluation

From the early days of the Internet, researchers have studied how people deploy heuristics—practical strategies or rules of thumb—to decide what websites to trust. Effective heuristics, defined by Gigerenzer and Gaissmaier (2011, p. 454), allow people to ignore "part of the information, with the goal of making decisions more quickly, frugally, and/or accurately than more complex methods." On the other hand, *weak heuristics* (McGrew, 2021), i.e., heuristics that rely on easily gamed surface features—such as a site's layout or URL—may seem like they save time but often lead to dubious conclusions (Breakstone, Smith, Wineburg, et al., 2021; Wineburg et al., 2016).

B. J. Fogg was among the first researchers to study how ordinary people deploy heuristics when judging websites (Fogg et al., 2001, 2003). In a study of 2,684 participants, 46% named site design as the top factor when explaining why they deemed one site more credible than another (Fogg et al., 2003).

Fogg's work set the tone for the first generation of heuristics-based Internet research. In a study with college students, Hilligoss and Rieh (2008) asked participants to record instances of information-seeking. Researchers identified four heuristics that students used to quickly locate and judge information. These were *genre* (e.g., privileging established websites over blogs); *source related* (determining whether a source was primary or secondary); *endorsement* (testimonials from friends or authorities); and *aesthetics* (or as one participant put it, “a nice and clean layout” p. 1471).

Visual appearance also loomed large in a study by Flanagin and Metzger (2007) in which 574 participants assessed websites across a variety of genres. A follow-up investigation (Metzger et al., 2010) focused explicitly on the heuristics participants used. From focus groups with 109 respondents, the researchers distilled five heuristics, two of which (reputation of a site and endorsement) overlapped with Hilligoss and Rieh (2008). In addition, the researchers identified “expectancy violation,” a mismatch between sites that claimed to be authoritative but had spelling errors and/or a poor layout; “persuasive intent” that steered users away from sites with ads; and the “consistency heuristic,” in which users “validated information by checking different Web sites to make sure the information was consistent” (Metzger et al., 2010, p. 428).

With the exception of the “consistency heuristic,” these heuristics are now largely ineffective because they train attention on a site’s internal features: its aesthetics, its ease of navigation, whether its links work, the presence of ads, and so on. The Web is not the same Web as it was twenty or even ten years ago. In the Internet’s early days, web design was a handy proxy for professionalism. Websites were costly to create, and amateurishness prevailed. But in an age of inexpensive Web templates, judging a site by its surface features is fraught with problems. Advertisers, lobbyists, and assorted bad actors create elegant sites that glitter with the surface features of credibility, lending an air of respectability to organizations that mask their real intent (Allcott & Gentzkow, 2017; Kozyreva et al., 2020; Sullivan, 2019). Despite vast changes to the Internet’s landscape, educational materials continue to offer advice that last held true when the Internet was accessed by a dial-up modem. Major universities still equate sites bearing a dot-org domain with nonprofit organizations, despite the fact that dot-org is an open domain available to anyone who pays a fee—including hate groups that bank on dot-org's perceived respectability (Wineburg et al., 2020; Wineburg & Ziv, 2019).

Students' Online Evaluations

Indeed, many studies have shown that students rely on weak heuristics to make judgments of credibility. Across continents and educational contexts, students are hoodwinked by such features as a site's look, top-level domain, objective-sounding language on its About page, and links to prestigious outlets (Bakke, 2020; Barzilai & Zohar, 2012; Breakstone, Smith, Wineburg, et al., 2021; Dissen et al., 2021; Gasser et al., 2012; Hargittai et al., 2010; Lurie & Mustafaraj, 2018; Martzoukou et al., 2020; McGrew et al., 2018; Wineburg et al., 2020). Students often treat the order of search-engine results as a proxy for trustworthiness, a faulty "top link" heuristic (Pan et al., 2007). In the largest and most comprehensive study of its kind, Breakstone, Smith, Wineburg, et al. (2021), asked 3,446 high school students to evaluate websites using a live Internet connection. One site was co2science.org, which denies humans' role in global warming while claiming to "disseminate factual reports and sound commentary." A quick search reveals the group's ties to the fossil fuel industry. But 96% of students never uncovered this connection. Too often their evaluations stalled at three letters: "This page is a reliable source to obtain information from," one student wrote, "you see in the URL that it ends in .org as opposed to .com" (p. 509).

Students' problems start before they make their first click. Reviewing search results, students tend to float aimlessly across the screen, "touching or not touching pieces of information . . . unconscious to its value and without a plan" (Kirschner & van Merriënboer, 2013, p. 171). Despite a reliance on faulty strategies, many people retain a steely confidence in their Web savvy. A review of 53 studies of information seeking (Mahmood, 2016) showed the

Dunning-Kruger effect in full force. People overestimate—in many cases, wildly overestimate—their ability to discern online fact from fiction.

How do Experts Navigate the Web?

A smaller set of studies has explored how experts evaluate digital content. Lucassen and Schraagen (2011) compared an "expert" group, people who either worked in the auto industry or were car enthusiasts active in online forums, with novices. People who knew about cars were better able to detect errors in Wikipedia than people who didn't.

In another study, Brand-Gruwel and colleagues (2005) designated a group of five PhD students in educational technology as experts and compared them to a group of first-year psychology students. Participants thought aloud as they searched the Internet to answer questions about the perishability of food. Based on think-aloud protocols, the researchers divided online problem-solving into four categories: orientation, monitoring, steering, and testing. The researchers described a subcategory, "judge scanned information," as "Judge[d] the reliability of sources and the quality and relevance of information" (p. 504). Left unspecified was how people did this.

In a research program that took up this issue, Wineburg and McGrew (2016, 2017, 2019) compared three groups of experienced Internet users as they judged the credibility of unfamiliar websites: undergraduates from Stanford University, professional historians from five different universities, and professional fact checkers at the nation's most prestigious news outlets. Fact checkers were included in the sample because they are information generalists who in a single workday navigate topics that traverse a wide swath of disciplines, topics, and research methodologies (Godel et al., 2021; Graves, 2017; Silverman, 2020).

Historians and college students fell victim to weak heuristics, such as official-looking logos and domain names. They approached websites *vertically,* reading from top to bottom, spending time examining features within a site. In contrast, fact checkers deployed the heuristic of *lateral reading* (Wineburg & McGrew, 2017, 2019), a term that draws on the image of multiple browser tabs arrayed across the horizonal axis of the screen. Instead of first examining a site's internal features (which are, after all, controlled by its designers), lateral readers evaluate unfamiliar sites by *leaving* them and turning to the open Web. By reading laterally, fact checkers reached more warranted conclusions in less time than the other two groups (Wineburg & McGrew, 2019).

Evaluation of Sources in Print and Digital Environments

Research on how people vet information obviously predates the Internet. In one of the first studies of how readers integrate information across multiple sources, Wineburg (1991) compared professional historians to a group of talented high school students as they read eight sources. Documents were carefully selected, curated, and excerpted to highlight issues open to interpretation. Historians, but not students, zoomed in on a document's bibliographic information: Who wrote it and when, its genre (e.g., sworn testimony versus a diary entry versus a textbook entry), and the relation of the authors to the events they report. Wineburg dubbed this orientation to the text as *sourcing*:

> By knowing a document's author and the place and date of its creation, historians could develop what would be in the body of the document, the stances it might take and its truthfulness or accuracy. . . . The sourcing heuristic . . . provid[es] an anticipatory framework for the subsequent encoding of text (p. 79)

This early work proved generative to researchers studying the nature of historical interpretation (e.g., Freedman, 2015; Nokes et al., 2007; Rouet et al., 1997; Stahl et al., 1996; Wiley & Voss, 1996), as well as to investigators who extended multiple source research to science (e.g., Brante & Strømsø, 2018; Bråten et al., 2009; Goldman et al., 2012; Stadtler et al., 2013; Strømsø et al., 2013; Wiley et al., 2009). The efflorescence of research on multiple source reading has yielded influential models that describe how readers integrate information across texts of varying accuracy and authority.

One of the best-known is Rouet and Britt's (2011) MD-TRACE (multiple-document task-based relevance assessment and content extraction). According to this model, readers follow a series of steps: they first set goals by reading task instructions and formulating a plan drawing on domain knowledge and experiences with similar tasks. They then decide whether to consult additional resources by assessing prior knowledge (e.g., Do I know enough to answer this question?), as well as the task directions (e.g., Do directions say to consult external resources?). In light of the task demands, they decide which sources matter, read and integrate content across multiple sources, and use what they learned to respond to the task. Finally, they decide whether their product meets the goals they set at the task's outset; if not, they return to previous steps until they have sufficiently improved the product.

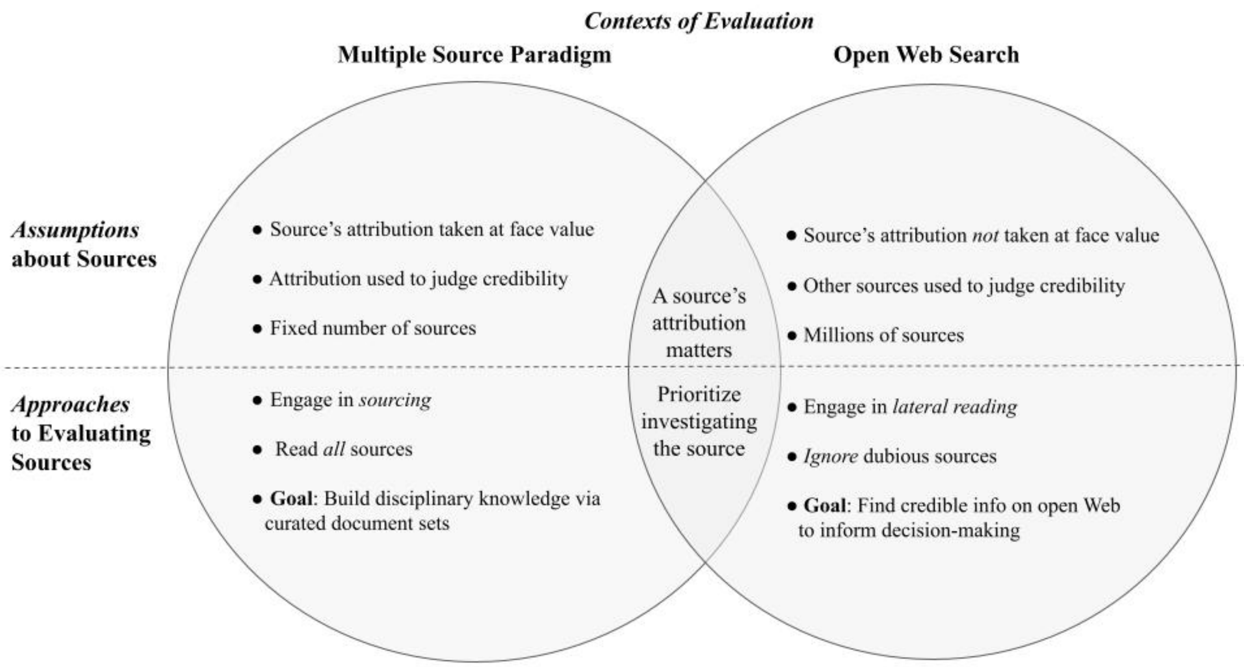
MD-TRACE and similar models make few distinctions between reading print and digital sources. Indeed, the "multiple source" paradigm (Braasch et al., 2018) is deeply rooted in the assumptions of an analog world. (See Kammerer et al., 2021, for an important exception.) Researchers typically restrict access to the Web by selecting, excerpting, and curating closed document sets, even though the Web contains two billion sites that can be explored in infinite ways. For example, a study that set out to investigate students' "evaluation behaviour while searching the internet" culled eight Internet sources, printed them out, and presented them to students in a booklet (Walraven et al., 2013). Another investigation studied how students "validate online documents" but presented them in an offline browser with summaries of the original (Mason et al., 2014). Similarly, Pérez and colleagues had students "search for information" on a paper version of a Web page and indicate on a Likert scale the "links" they would most trust (2018, p. 57). Other times, researchers pull documents from the Web but edit them so they are equivalent in length (e.g., Salmerón et al., 2013).

When students fail to pay attention to source information, researchers sometimes alter the attribution to boost its salience while leaving the document's contents intact [e.g., changing an author from a "sales-manager computer engineer" to a "secondary school student" so that students would better "detect the author competence issue" (Macedo-Rouet et al., 2019, p. 312)]. Researchers have deployed various strategies to simulate a digital environment: "Google-like" interfaces that remain offline (Mason et al., 2014; Strømsø et al., 2013); paper sources adapted to look like they came from the Web (Macedo-Rouet et al., 2019); a Search Engine Results Page limited to preselected (and adapted) sites (Goldman et al., 2012); and an intranet with hyperlinks disabled (Salmerón et al., 2013).

These adaptations strike a Mephistophelian bargain: They sacrifice ecological validity for experimental control. They abjure the Internet's most basic attribute: providing access to a galaxy of electronically-linked sources that can be searched via infinite pathways. The Web is a web; it is not a document set. (See Figure 1.)

Figure 1

Multiple Source Paradigm Versus Open Web Search



An even greater chasm exists between the conceptualization of sourcing in the multiple source paradigm and what that concept means on an Internet teeming with sites that shrewdly conceal their real backers (McHenry, 2018). When participants in a multiple document study examined, for example, a letter by Abraham Lincoln (Wineburg, 1998) or an article on climate change by "an expert in sustainable energy" (Macedo-Rouet et al., 2019), their task was to coordinate the attribution with the document's contents to assess "the trustworthiness of the information in light of the characteristics of the source" (Bråten et al., 2009, p. 7). Never was the goal to question whether Lincoln's letter might be a forgery or whether the "expert in sustainable energy" fudged her credentials. Researchers sign an unwritten pact with subjects that the source is who and what it says it is.

It is precisely this feature—the act of taking bibliographic information at face value—that causes trouble on the Internet (Jack, 2014; McHenry, 2018; Oreskes, 2019; Silverman, 2020). Consider how readers would be led astray were they to take at face value the source information on the polished website of the Employment Policies Institute (www.epionline.org). The About page touts the organization as "nonpartisan," and features research "by independent economists at major universities." The research tab provides downloadable reports from scholars at universities like Carnegie Mellon, San Diego State, and the University of New Hampshire. (Scholars at these institutions did write these reports; sometimes with funding provided by the Employment Policies Institute.) The site sports a dot-org top-level domain and carries a 501(c)(3) tax-exempt status from the U.S. Internal Revenue Service. A reader dwelling on the site, however, would never learn that this "nonpartisan" organization is run out of a Washington, D.C., public relations outfit that specializes, according to *The New York Times*, in creating "official-sounding nonprofit groups to disseminate information on behalf of corporate clients" (Lipton, 2014). The only way to quickly learn about the true nature of this source is to leave the site and search for information about the organization on the open Internet.

Conceptual Underpinning

We set out to design a curriculum that would help students make quick but accurate decisions on the open Web. Our thinking was informed by Herbert Simon's idea of *bounded rationality* (1990). Skilled problem-solving in a digital environment demands the coordination between the awesome processing capacities of computer technologies, on one hand, and the limitations of beings whose processing capacities are easily overwhelmed, on the other. As Simon (1971) explained at the dawn of the information age:

> The wealth of information means a dearth of something else: a scarcity of whatever it is that information consumes. . . . It consumes the attention of its recipients. Hence a wealth of information creates a poverty of attention and a need to allocate that attention efficiently among the overabundance of information sources that might consume it (pp. 40-41).

Simon's insights provided the backdrop for the pioneering work of Gerd Gigerenzer and his associates at Berlin's Max Planck Institute (Gigerenzer, 2022; Gigerenzer & Gaissmaier, 2011). In a research program that explores how professionals make decisions under conditions of uncertainty, Gigerenzer and Gaissmaier (2011) have identified scores of *fast and frugal* heuristics that allow practitioners to arrive at outcomes more quickly and accurately than more time-consuming methods. For example, in criminal profiling, police have relied on complicated models to predict where a serial offender is most likely to live. They consider multiple inputs to predict probabilities. A fast and frugal alternative that bested more complex methods is the "circle" heuristic, in which a circle is drawn around the two farthest-flung crime locations and predicts that the offender will live in the center (Snook et al., 2004). More information does not always guarantee better decision-making.

Indeed, fact checkers employed a series of heuristics that allowed them to read less but learn more than other intelligent adults. In designing our curricular approach, we drew on the heuristics used by professional fact checkers (Wineburg & McGrew, 2019). We posited that three related constructs were responsible for their success: *footing, taking bearings,* and *lateral reading* (see Figure 2).

Figure 2

Fact Checkers' Approach to Digital Information

Footing refers to the understanding of the Internet a user brings to online searches: how one plants one's feet before typing the first word into a browser. Fact checkers bring to search an understanding of the difference between a medium in which anyone can post anything without obtaining permission and traditional print environments, in which books, articles, and research reports are vetted by multiple gatekeepers. Fact checkers understand how Web crawlers index websites and how to choose unique keywords accordingly; they discount weak heuristics, aware of their susceptibility to manipulation; they possess a working knowledge of search engine optimization and how it affects search order; and they understand how inputs tilt search in predetermined directions and can lure users to data voids exploited by bad actors (Golebiewski & boyd, 2019; Tripodi, 2020).

Taking bearings is honing a sense of direction when starting an Internet search. Experienced hikers take bearings in a foggy terrain. They rotate their compass's bezel to determine bearings—the angle, measured in degrees, between north and a fixed point—and proceed apace toward their destination even when fog diminishes their visibility. Obviously, Web searchers don't use a physical compass. Instead, they apply fast and frugal heuristics that allow them to get a quick lay of the land. Principle among them is *click restraint:* holding back the urge to click a search result until one surveys the "information neighborhood" (Klurfeld & Schneider, 2014). Fact checkers view the search engine results page (SERP) as a gestalt that reveals who controls the digital prime real estate: the three top results account for over half of all clicks (McGrew, 2021; Shelton, 2017). Click restraint enables *result mining,* the act of extracting clues from the SERP's snippets (the brief excerpts that accompany each result), and *critical ignoring*, passing over low-quality results, thus increasing the chance that the first click will be a wise one (Wineburg, 2021).

Lateral reading is the act of leaving an unfamiliar website to search the open Internet. The opposite of *vertical reading,* lateral reading is the operational arm of taking bearings. Fact checkers leave an unfamiliar site in seconds, opening new tabs across the horizonal axis of their screen to investigate the organization or individual behind the original site. Under conditions of attention scarcity (Simon, 1990), lateral reading postpones an investment of attention until first determining that such an investment merits the effort (McGrew, 2021; Wineburg, 2018, 2021).

Present Study

The present investigation builds on a small but promising body of research showing that brief interventions, based on strategies used by professional fact checkers, can improve students' judgments about the reliability of digital sources. In a study across three high school history classes, students improved from pre- to posttest after eight lessons over two months (McGrew, 2020). College students in both synchronous (Brodsky, Brooks, Scimeca, Galati, et al., 2021; Brodsky, Brooks, Scimeca, Todorova, et al., 2021; McGrew et al., 2019) and asynchronous courses (Breakstone, Smith, Connors, et al., 2021) improved in their ability to judge digital content—in some cases in as little as 150 minutes of instruction (McGrew et al., 2019). The present study is the first to implement a curriculum intervention across an entire urban school district. It asked whether high school students would improve as evaluators of online content on the open Web after completing six 50-minute lessons taught by their teachers.

Methods

Curriculum Framework

Principles of *cognitive apprenticeship* (cf. Collins et al., 1989; Dennen, 2004) guided our curricular approach. Five of the six lessons featured *cognitive modeling*, an instructional program that makes expert cognitive processes visible to novice learners (Brown et al., 1989; Collins et al., 1991; De La Paz et al., 2016). In two lessons, teachers directly explained Civic Online Reasoning strategies for evaluating online content and modeled them with sources from the Internet. In three additional lessons, teachers articulated their thoughts (an instructional think-aloud) as they evaluated online content so that students could see how expert strategies are applied in real time.

Lateral reading was at the core of our approach. The curriculum required students to evaluate genuine, unaltered online sources using a live Internet connection. During the first lesson, teachers projected their computer screens to show students how to approach minimumwage.com, a site purporting to offer nonpartisan information about minimum wage policy (see Table 1). Afterwards, students practiced lateral reading with an article about minimum wage policy from theodysseyonline.com. By opening up a new tab and searching for "Odyssey Online website," students found sources like a CNBC article titled, "Thousands of college kids are behind a 'clickbait' publishing platform" (Porter, 2017). Students were provided guiding questions and graphic organizers to support their thinking (Collins et al., 1989; Monte-Sano et al., 2014a, 2014b). Each lesson included opportunities for students to share their reasoning so teachers could monitor student thinking and provide support.

Table 1

Modeling of Lateral Reading

Instructional Sequence
Step 1: Teachers began by projecting their screen as they navigated to minimumwage.com. Teachers "thought aloud" as they evaluated the site. Although the site is professionally designed, teachers explained that the most important consideration in evaluating a website is determining who is behind it.
Step 2: Teachers navigated to the About page, which indicated that minimumwage.com is a project of the Employment Policies Institute, a "non-profit research organization dedicated to studying public policy issues surrounding employment growth." Teachers explained that although students should not base their evaluation of a site on the content of its About page, the About page can provide useful information for lateral reading.
Step 3: Teachers explained that they were unfamiliar with the Employment Policies Institute and would read laterally. They opened new tabs in their browsers, searched for Employment Policies Institute, and modeled how to comb through the search engine results page (SERP) looking for sources that might shed light on the organization.
Step 4: Teachers clicked on major new sources. An article from *The New York Times* identified Berman and Company as the organization behind the Employment Policies Institute and described the public relations firm's work on behalf of corporate interests. Teachers ended the lesson by recapping how to engage in lateral reading.

Setting

This study took place in an urban Midwestern United States school district with over 40,000 students. Nearly half qualified for free or reduced-price lunch. One third were racial or ethnic minorities, and 7% were English language learners.

Participants

Students

Participants were students enrolled in a district-required government class at each of the district's six high schools in spring 2019. All students (n = 762) were invited to participate in the pretest and posttest, and 545 completed both outcome measures. Students from three high schools completed experimental lessons (n = 289); students from the other three high schools (n = 256) served as controls.

Schools

Schools were matched based on race/ethnicity and the percentage of students enrolled in the free/reduced lunch program. Matched schools were assigned to opposite conditions (see Table 2).

Teachers

All of the districts' government teachers (n = 12) participated in the study. Teachers were assigned to treatment condition at the school level. Teachers at three high schools were assigned to the treatment condition (n = 6, two teachers per school), and teachers at the other three schools were in the control condition (n = 6, two teachers per school). Because teacher collaboration in the district was primarily within the same school, assigning teachers at the school level decreased the chance that teachers would communicate or collaborate across condition. District administrators and the research team also requested that teachers refrain from sharing the treatment curriculum materials until the end of the semester.

Teachers in the treatment group had an average of 17 years of teaching experience (*SD* = 14.2; *Mdn* = 12; range: 5 to 42) and had been at their school for an average of 16 years (*SD* = 15; *Mdn* = 11; range: 3 to 42). Teachers in the control condition averaged 16.3 years of teaching experience (*SD* = 5.7; *Mdn* = 18; range: 9 to 22) and had been at their school for 11.3 years (*SD* = 6; *Mdn* = 12.5; range: 9 to 17). All teachers were certified to teach social studies and held a BA in either history or social studies education. Five of the six treatment teachers and all six control teachers held a master's degree in history or education. Prior to the intervention, none of the teachers reported being familiar with the Civic Online Reasoning curriculum (cor.stanford.edu) or had received professional development about the approach.

Table 2

Treatment and Control Schools

	Treatment	Control
Enrollment	6576	5966
Race/Ethnicity		
American Indian/Alaska Native	1%	0.7%
Asian/Pacific Islander	5.2%	4.7%
Black/African American	5.9%	6.5%
Hispanic/Latino	12.9%	14.7%
Two or more races	7.4%	7%
White	67.6%	66.4%
Free/reduced lunch participation	39.3%	45.2%
Female	49.2%	48%

Intervention

Over three months, treatment teachers integrated six 50-minute experimental lessons into their regular curriculum (see Table 3). Teachers received detailed, classroom-ready materials for students and teachers (see Appendix A[1] for a sample lesson plan). Directions included sequences of instruction, discussion prompts, and possible responses to student misconceptions. Some lessons included slides for teachers to project. Student materials, delivered via Google Docs and Forms, included graphic organizers, guiding questions, and links to sources to evaluate. To test the efficacy of the intervention, we first piloted it with more than 400 students in an urban school district on the West Coast of the United States (see Appendix B).

Table 3

Civic Online Reasoning Lessons in the Intervention

Lesson	Summary of activities
Introduction to lateral reading	1. Students independently evaluated an article from minimumwage.com, a site that obscures its purpose and backers. 2. Teacher introduced the importance of investigating online sources and modeled how to invest attention wisely by reading laterally about minimumwage.com. 3. Students, in small groups, practiced reading laterally. They investigated a crowd-sourced website (theodysseyonline.com) that posted an article about minimum wage policy. 4. Using an article about the minimum wage as a starting point, students independently searched online to learn about *The Seattle Times*, Seattle's main newspaper.
Lateral reading resources and practice	1. Class reviewed lateral reading as a strategy for vetting online sources; teacher introduced Wikipedia as resource to use when reading laterally. 2. Students practiced reading laterally about a fake news website, first by going to a Wikipedia entry and then examining its accompanying references. 3. Teacher introduced fact-checking sites (e.g., Snopes, PolitiFact) and established news organizations as additional resources to use when reading laterally. 4. Students practiced reading laterally by contrasting the Brookings Institution, a reputable Washington, DC, think tank, with the Center for Immigration Studies, a partisan group that advocates against immigration.

[1] Appendices A-D can be found in the online supplemental materials.

Lesson	Summary of activities
Vertical vs. lateral reading	1. Class compared what they learned about the Against Malaria Foundation (againstmalaria.com) when reading vertically (i.e., dwelling on the site and clicking the site's internal links) versus reading laterally by leaving the site. Teacher modeled how giving weight to superficial features (e.g., the site's dot-com domain or its spare, bare-bones design) could lead to the erroneous conclusion that againstmalaria.com and the organization behind it are unreliable. 2. Students discussed differences in what they learned about Against Malaria from a vertical versus a lateral read. Teacher reminded students that investigating the source by reading laterally led to better conclusions. 3. Teacher modeled how easily one can become misled by reading a social media post about a medical treatment for malaria from Dr. Joseph Mercola, a self-proclaimed "health activist," who has been censured by the Food and Drug Administration. 4. In small groups, students searched online to investigate Mercola and discussed differences in what they learned from lateral versus vertical reading. Teacher emphasized that information from unvetted sources cannot be accepted at face value and that students should not invest time in information from questionable sources.
Analyzing digital evidence	1. The class watched a video from Twitter that purports to show a protest in Iran. The footage actually comes from Bahrain. Students collectively brainstormed questions they would ask about the video. 2. Teacher explained the importance of only investing attention in evidence from reliable sources. 3. Students worked in small groups to analyze additional examples, including a YouTube video, tweets, and an infographic. They discussed their evaluations as a whole class. 4. Students independently analyzed evidence from an anonymous user in an online discussion forum.
Social media verification	1. Teacher gave an overview of skilled approaches to verifying claims on social media and modeled using these approaches to verify a claim made by a freelance journalist on Twitter. 2. Students worked in small groups to practice verifying Facebook and Twitter claims from various sources, ranging from a liberal political advocacy group to a *Los Angeles Times* reporter. 3. Students completed a formative assessment that asked them to analyze a misleading tweet about gun ownership.

Lesson	Summary of activities
Click restraint	1. Teacher explained how search results are organized and how to examine the main elements of the search engine results page (SERP). Topics included: search algorithms, search engine optimization, page snippets, and top-level domains (in particular, that anyone can purchase a dot.org domain without going through a vetting process). 2. Teacher introduced *click restraint:* scanning the SERP, mining snippets to look for recognizable sources, and making an informed decision about where to click first. 3. As a whole class, students analyzed a list of search results about the pros and cons of Internet filters in schools; they discussed which links they might select and why. 4. Students worked in small groups to practice click restraint when reviewing search results about school uniforms and then presented findings to the whole class.

Teachers filmed themselves teaching each of the six lessons. A member of the research team viewed each lesson and rated the fidelity of implementation (see rubric, Table 4). Across 34 videotaped lessons (two files were corrupted), fidelity scores averaged 3.56 (SD = .56) on a 4-point scale. A second rater scored 12 videos (35% of the data corpus). Interrater agreement was 91.7% (Cohen's *kappa* = .82).

Table 4

Lesson Plan Fidelity Rubric

Fidelity Score	Description
0	Most elements of the lesson plan were missing or substantially modified.
1	Some aspects of the lesson were taught, but more than one element was missing and/or multiple activities were substantially modified.
2	Almost all aspects of the lesson were taught, but one element was missing (e.g., one round of guided practice was skipped) *and* one activity was substantially modified (e.g., what was supposed to be a full-class demonstration was instead individual practice time for students).
3	The lesson was taught with only minor modification (e.g., students worked individually instead of in groups during guided practice), but one element was missing (e.g., one round of guided practice was skipped) or one activity was

	substantially modified (e.g., what was supposed to be a full-class demonstration was instead individual practice time for students).
4	All aspects of the lesson, as outlined in the lesson plan, were taught with only minor modifications (e.g., students worked individually instead of in groups during guided practice).

Teacher Professional Development

Teachers in the treatment group attended a day-long professional development workshop. The research team introduced the Civic Online Reasoning curriculum and provided teachers with six lesson plans. After teaching the second and fourth lessons, teachers consulted virtually with members of the research team.

Teachers in the control group taught the normal government curriculum, which included a study of founding political documents, topics in political science, and analyses of current events. They completed a survey about their teaching practices, submitted their course syllabi, uploaded six lesson plans representative of the content of their courses, and administered pre- and posttests to their students. To increase control teachers' engagement with the project and to provide equitable access to the curricular resources, control teachers attended a day-long professional development workshop about the Civic Online Reasoning curriculum after the posttest administration. This session matched what treatment teachers received at the beginning of the study. Both groups of teachers received identical stipends as tokens of appreciation.

Assessments

Students in both conditions completed a pretest at the start of the semester and a posttest at the end. Both pretest and posttest included seven constructed-response items designed to assess aspects of Civic Online Reasoning (cf. McGrew et al., 2018, 2019; Wineburg et al., 2016; see Appendix C for descriptions of each item). Teachers read a script provided by researchers, and students answered the items using the survey platform Qualtrics on laptop computers. Each item presented students with an online stimulus. Each assessment item took between three and eight minutes to complete.

The pretest and posttest items included a range of sources that broadly reflect the type of content students encounter when they go online. Students could search online at any time for every exercise in the assessment. However, not all items required students to conduct open Web searches. Some provided screenshots of sources rather than live links; it was possible to earn full credit on these tasks without searching online. Factoring into this design were several issues: First, the school district's Internet filter prevented students from directly accessing social media sites like Twitter and Instagram; screenshots provided a way to assess how students used the kinds of sources they encounter outside of school. Second, screenshots are frequently shared through social media, so these tasks reflected that reality. Finally, in cases where sources lack adequate context, it's often best to conclude that there's not enough information presented to make a good decision. For example, one task displayed a tweet that showed a child, purportedly in Syria, lying between mounds of dirt identified in the post as his parents' graves. In fact, the photo was taken in Saudi Arabia as part of an art project and did not actually depict a Syrian child; the supposed "graves" were piles of rocks (Traywick, 2014). To receive full credit, students needed to question either the source of the post (e.g., that it lacked information about the tweet's author) or the photograph (e.g., there was no proof the photo was taken in Syria or that the rocks were graves). Although some students did go online to locate articles debunking the tweet, they did not have to do so to receive full credit.

Other tasks required students to go online to evaluate claims. One item asked students to imagine they were researching global warming and landed on a website from an organization called “Friends of Science” (see Figure 3). Students were explicitly told they could use any online resource, including opening new tabs and doing Internet searches, to decide if friendsofscience.org was credible. In addition to evaluating the site’s trustworthiness, the task prompted students to explain their answers and cite evidence from other sites. To earn full credit, students had to look beyond the site’s appearance—including a dot-org URL, references to scientific studies, and staff with advanced degrees—and read laterally about the group behind the site. Students who completed these steps learned that the site, which argues that humans are not responsible for climate change, receives backing from the fossil fuel industry. [2]

The pretest and posttest used parallel forms to avoid testing effects that could have arisen had students completed the same items twice. Students in both treatment and control classrooms completed Form A at pretest and Form B at posttest. Parallel items presented similar directions and questions but different stimuli. For example, the parallel form of the climate change task presented an identical prompt (“Is this website a trustworthy source for learning about global warming?”; Figure 3) for evaluating a second site, co2science.org, which also contests the scientific consensus on climate change. The site’s sponsor, the Center for the Study of Carbon Dioxide and Global Change, is backed by ExxonMobil—a fact easily discovered by reading laterally.

Student responses were scored on a three-point rubric (Beginning - 0; Emerging - 1; Mastery - 2), a scoring approach our research group developed and tested in past projects (cf. McGrew et al., 2018). In *Mastery* responses, students demonstrated clear proficiency in the targeted reasoning processes. *Emerging* responses were either partially correct but incomplete, or reflected aspects of proficient as well as problematic reasoning. *Beginning* responses revealed

[2] Students could also receive credit for reasoning that the website was not credible because it presented information contrary to the scientific consensus about global warming.

Figure 3

Parallel Pretest and Posttest Items

Pretest Item

Please take about 8 minutes to complete this task.

You are researching global warming and come across this website: https://friendsofscience.org. Please decide if this website is a trustworthy source of information on global warming. You can open a new tab and do an Internet search if that helps.

Is this website a trustworthy source for learning about global warming?

Explain your answer, citing evidence from the webpages you used. Be sure to provide the URLs to the webpages you cite.

Posttest Item

You are researching global warming and come across this website: http://www.co2science.org. Please decide if this website is a trustworthy source of information on global warming. You can open a new tab and do an Internet search if that helps.

Is this website a trustworthy source for learning about global warming?

Explain your answer, citing evidence from the webpages you used. Be sure to provide the URLs to the webpages you cite.

incorrect or irrelevant reasoning about online sources. (See Table 5 for a sample rubric with student responses at each level of performance.) Students' writing quality did not factor into the scoring rubrics; it was possible for students to earn a Mastery score with a partial sentence as long as it reflected a clear understanding of the concept targeted by the task.

Table 5

Rubric and Sample Student Responses for Syrian Child Tweet Task

Rubric level	Level description	Sample student response
Beginning	Student argues that the post provides strong evidence or uses incorrect or incoherent reasoning	"It shows that some of the children have no parents and he or she is very sad to have no parents."
Emerging	Student raises questions about the post but does not consider its source or the origin of the photograph.	"There is no evidence to prove that those are the child's parents."
Mastery	Student argues that the post does not provide strong evidence and questions the source of the post (e.g., we don't know anything about the author of the post) and/or the source of the photograph (e.g., we don't know where the photo was taken).	"How do I know this picture is real? I don't know who posted it, how they obtained this photo, or if it was even shot in Syria. I just can't rely on this image alone."

Two raters scored each task for both the pretest and posttest. Scoring was blind to condition. Interrater reliability was high for both pretest and posttest, with weighted *kappa* estimates above .90 for each task. (See Appendix D for weighted *kappa* coefficients.) Cronbach's *alpha* was used to estimate internal consistency of the pretest ($\alpha = .58$) and the posttest ($\alpha = .69$).

Analysis

To account for the nested nature of the data, we used a multilevel linear mixed model to test the difference in pre- to post-scores. We tested whether students in the treatment group showed greater evidence of learning from pretest to posttest than control students. Random assignment occurred at the school level, with three schools assigned to the treatment group and three to the control. Unlike participants in fully randomized designs, students in school-based research are often not completely independent but are nested (or clustered) in classrooms and schools.

Teachers (two per school) were nested within schools (n = 6), and students (n = 545) were nested within teachers. A sequence of four regressions were run to determine the model of best fit:

> **Model I** included fully crossed fixed effects for *time* and *treatment* and random effects for *students* (nested in schools), *teachers* (nested in schools), and *schools*. Intercepts for random effects were free to vary according to a normal distribution, and slopes were fixed. This model had a marginal R^2 of .15 and a conditional R^2 of 0.67. In other words, the fixed effects explained 15% of the variance while the combination of the fixed and random effects explained 67%. Results revealed that schools accounted for only 1% of the variance in observed scores, so we eliminated random school effects in Model II.

> **Model II** continued to treat intercepts of the random effects for *students* (nested in teachers) and *teachers* as varying according to a normal distribution. Model II had a marginal R^2 of .15 and a conditional R^2 of 0.66. A Chi-square test of difference in the fit for the two models confirmed that dropping the school factor did not have a significant impact [$X^2(1) = 0.369$, $p = 0.876$].

To account for potential differences in how teachers implemented the Civic Online Reasoning curriculum, **Model III** allowed the slopes for *teachers* (a random factor) to vary. This adjustment improved model fit [$X^2(2) = 14.20$, $p < 0.001$] and had similar R^2 values (marginal $R^2 = 0.15$; conditional $R^2 = 0.68$).

Model IV added the following fixed-effect covariates: race, ethnicity, gender, primary language spoken at home, hours spent online per day, and how frequently students checked whether online information was trustworthy. Adding race, ethnicity, and language controlled for variables relevant to educational outcomes. The addition of hours spent online and frequency of checking information controlled for potentially relevant online behavior. This model had a marginal R^2 of .23 and a conditional R^2 of 0.66. In other words, the fixed effects now explained 23% of the variance and the combination of the fixed and random effects explained 66%.[3]

Q-Q plot analysis of the residuals revealed outliers at the tails of the distribution that could affect the standard error of the parameters and signaled potential violations of the assumption of normality. We thus calculated robust parameter estimates produced with the *R* package *robustlmm* (Koller, 2016) to reduce outlier contamination, controlling for normality violations. Using squared residuals based on predictions from the robust parameter estimates, a test of the equality of variance across groups was non-significant, $F\ (1, 1224) = 3.10$, $p = .08$.

[3] Models I-III had the same fixed effects, which allowed Chi-square goodness-of-fit comparisons. Model IV added new fixed effects as covariates, so a comparison of fit with other mixed-effects models was not appropriate. We estimated fit for Model IV with marginal and conditional R^2 values. See Nakagawa and Schielzeth (2013) for why mixed effects models require the specification of marginal and conditional R^2 values.

Results

Four hundred and ninety-nine students with complete data for the covariates were included in the model for the final analysis. Table 6 shows the descriptive statistics for the variables used in the regression analysis by treatment and control groups.

Table 6

Descriptive Statistics by Control and Treatment Conditions

	Control (n = 228)		Treatment (n = 271)	
	Mean	*SD*	*Mean*	*SD*
Pretest score (out of 14)	2.27	2.05	2.93	2.49
Posttest score (out of 14)	2.83	2.35	5.06	3.13
Hours online per day	4.46	1.85	4.36	1.91
Frequency of checking trustworthiness (1 = Not Often, 5 = Always)	3.74	0.83	3.76	0.92
	Proportion	*SD*	*Proportion*	*SD*
Home language is English	0.88	0.33	0.89	0.31
Hispanic/Latino	0.14	0.34	0.10	0.30
Race				
White	0.69	0.46	0.64	0.48
Asian	0.08	0.28	0.10	0.30
Black or African American	0.05	0.21	0.11	0.31
Other race	0.04	0.19	0.05	0.22
More than one race	0.08	0.28	0.11	0.31
Female	0.52	0.50	0.47	0.50

Students in the treatment condition (*n* = 271) were more likely to show improvement from pretest to posttest than control students (*n* = 228), Robust $b_{\text{condition x time}} = 1.66$, $SE = .44$, $t = 3.77$, 95% CI (0.44, 3.77). As shown in Table 7, the Condition by Time interaction (bolded) was significant, while the main effects for Time and Condition were not. These results directly address effectiveness of this curriculum intervention.

Several covariates in Model IV were also significant predictors of student performance (see Table 7). English as the primary language spoken at home and self-reported frequency of checking the trustworthiness of online information were both positive predictors of performance. Identifying one's race as Black/African American or "other" was associated with lower performance as compared to White-identified students. Race/ethnicity was not a significant predictor for students who identified as Asian, Hispanic/Latino, or multiracial. Gender and the number of hours spent online per day were not significant predictors. Figure 4 shows the parameter estimates reported in Table 7 and 95% confidence intervals for each covariate in Model IV. Figure 5 reflects the estimated change in student scores from pretest to posttest after adjusting for all covariates in the model. Students' estimated scores in the treatment condition improved significantly more than the scores for students in the control group.

Limitations

We coped with multiple challenges in this intervention. Teachers were tasked with teaching content that only weeks before had been unfamiliar to them. Students, for their part, were thrust into an assessment situation that was similarly novel. Let loose on the open Web to evaluate the credibility of sources, they had to explain their reasoning using Qualtrics, an unfamiliar online survey platform.

We have little idea of the durability of student gains or whether they would practice the

Table 7

Fixed Effects Unstandardized Parameter Estimates and Robust Parameter Estimates for the Mixed Model Analysis

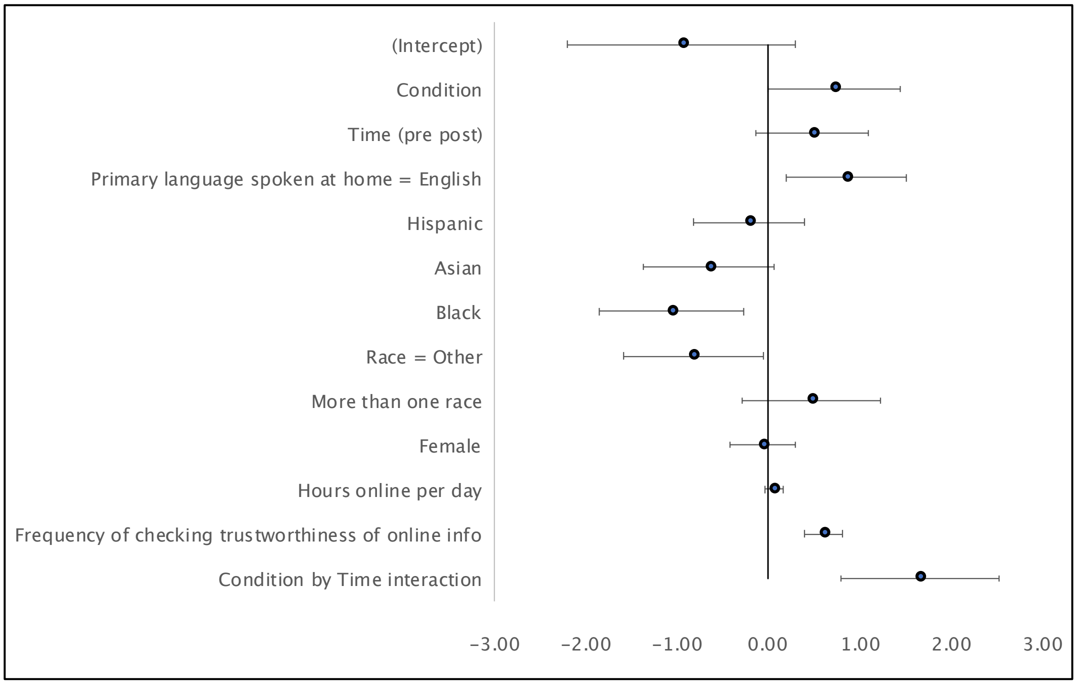
	Unadjusted parameter estimates			Robust parameter estimates		
	b	*SE*	*t*	*b*	*SE*	*t*
(Intercept)	-1.04	0.67	-1.56	-0.95	0.63	-1.51
Condition	0.70	0.46	1.50	0.71	0.37	1.95
Time (pre post)	0.53	0.28	1.87	0.48	0.31	1.52
Primary home language = English	0.88**	0.33	2.66	0.85	0.33	2.55
Hispanic/Latino status	-0.22	0.31	-0.73	-0.22	0.31	-0.71
Asian	-0.57	0.36	-1.58	-0.65	0.36	-1.79
Black	-0.93*	0.40	-2.32	-1.06	0.40	-2.68
Race = Other	-0.80*	0.39	-2.03	-0.82	0.39	-2.12
More than one race	0.37	0.38	0.98	0.47	0.38	1.23
Female	-0.04	0.18	-0.23	-0.07	0.18	-0.36
Hours online per day	0.08	0.05	1.56	0.06	0.05	1.22
Frequency of checking trustworthiness	0.62***	0.10	5.91	0.61	0.10	5.82
Condition by time interaction	**1.61****	**0.39**	**4.11**	**1.66**	**0.44**	**3.77**

Note. b = unstandardized parameter estimates; *SE* = standard error; Calculating degrees of freedom for robust parameter estimates is beyond the capability of statistical methods used to adjust for outliers. As a result, we report the robust t-values and show the associated 95% confidence intervals using the robust standard errors in Figure 4.

*p < .05, ** p < .01, *** p < .001

Figure 4

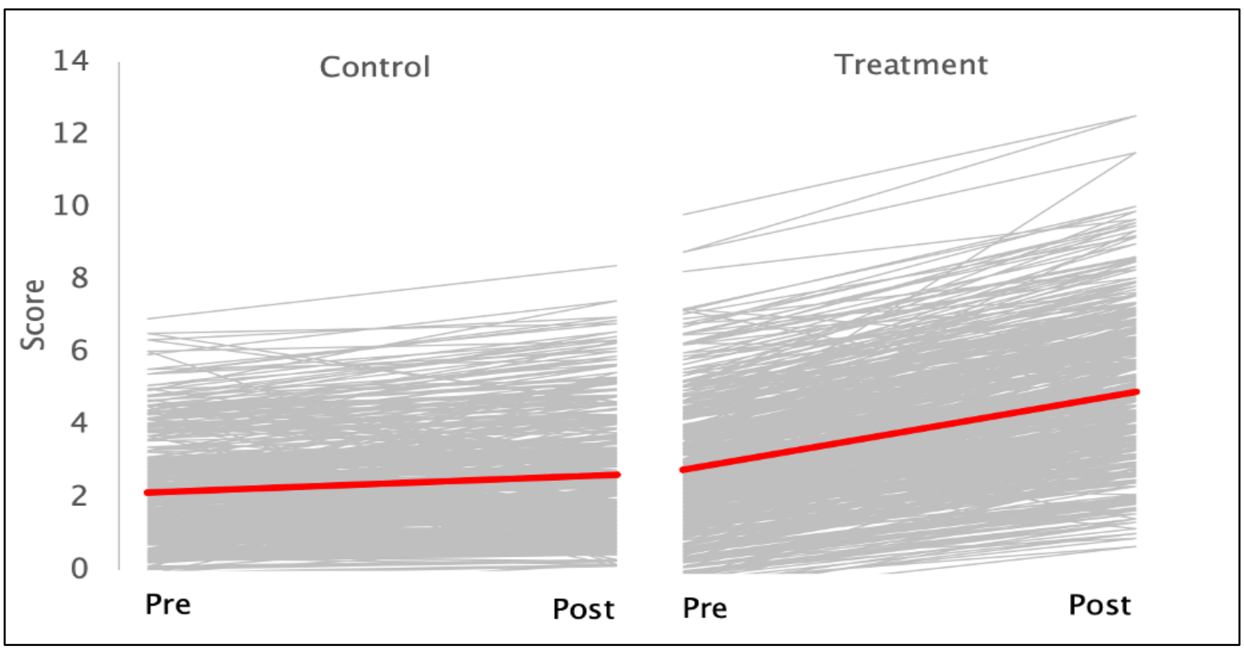
Robust Unstandardized Regression Coefficient Estimates with 95% Confidence Intervals



Note. 95% confidence intervals were constructed using the robust standard errors reported in Table 7.

Figure 5

Estimated Change in Student Scores from Pretest to Posttest After Adjusting for All Covariates



strategies of Civic Online Reasoning on their own. No transfer measures exist that gauge online reasoning. Nor do we know how students would tackle questions of their own choosing rather than the ones we provided. Furthermore, we lacked data about students' political beliefs and content knowledge, or their emotional responses to online content included in lessons and assessments, potential confounds that may have shaped students' responses. This concern, however, was mitigated by the fact that the same issues applied to students in control classrooms.

Finally, we relied on measures created by the research team because there were not suitable alternatives. One alternative would be to develop a series of intranet-based assessments. Although using an intranet would allow researchers to control the content students encounter, it comes at the cost of ecological and external validity. A closed intranet is not a realistic approximation of the open Web, and the moment the bell rings, students' phones access an Internet swarming with unvetted sources. Together these challenges shine a light on the substantial research that lies ahead.

Discussion

Dubious information spreads at alarming rates on the unregulated Internet, and no segment of the population is immune. Yet, the myth of digital natives, young people supposedly wise to the tactics that dupe their parents (Choi, 2020) holds on with the tenacity of a deeply-rooted blackberry bush. In our prior work, (Breakstone, Smith, Wineburg, et al., 2021), 3,446 high school students were provided with a live Internet connection and asked to evaluate a Facebook video that allegedly showed vote tampering during the 2016 Democratic primaries. The video's provenance was easily established on sites like Snopes or the BBC. Only three students in more than 3,000—less than one tenth of one percent—divined the true source of the video, which had, in fact, been shot in Russia.

Seen in this light, the present study provides cause for optimism. A modest intervention—just six 50-minute lessons—led to improvements in students' performance on measures of online reasoning. Our curriculum removed students from an "epistemically safe" environment (Chinn et al., 2020), where source information is readily available and assumed accurate, and exposed them to the wiles of a medium where sources may or may not be what they seem. The pretest and posttest assessments reflected the kinds of evaluations that young people need to make as they wade through their social media feeds. Each item asked students to evaluate real Internet sources in roughly five minutes per task—far shorter than typical school assignments but far longer than what people spend on "just in time" searches (Liu et al., 2010; Nielsen, 2011).

The implementation of this intervention reflected the realities of schooling and how programs are typically introduced in large districts. Teachers attended a one-day workshop and then used the curriculum materials in their classrooms. All government teachers, rather than a select group, participated. Our findings contribute to a growing body of evidence that shows that students can improve their online reasoning by learning a few easy-to-apply heuristics drawn from research with professional fact checkers. Positive results have accrued across age levels: middle school (Kohnen et al., 2020), high school (Axelsson et al., 2021; McGrew, 2020; Pavlounis et al., 2021), college (Breakstone, Smith, Connors, et al., 2021; Brodsky, Brooks, Scimeca, Galati, et al., 2021; Brodsky, Brooks, Scimeca, Todorova, et al., 2021; McGrew et al., 2019), and adulthood (Panizza et al., 2021).

At the same time, the results of the present study are not cause for unbridled celebration. Although students in treatment classes nearly doubled their scores from pre- to posttest, they still earned less than half of the available points on the assessment—an assessment that asked them, among other things, to evaluate the website of a not-for-profit organization funded by corporate interests and to compare an amply footnoted Wikipedia entry to a partisan source that carried the enticing allure of a dot-edu domain.

It would be unrealistic to expect an intervention of six 50-minute classes to do magic. The average American teenager spends more time online in one day—7 to 8 hours according to a survey by Common Sense Media (Rideout & Robb, 2019)—than the total time allotted to this intervention. Dislodging students' online habits, built over thousands of hours, will require a heavier lift.

The challenges of doing so are formidable. From the earliest years of schooling, students are provided texts that have been vetted by publishers, subject matter experts, adoption committees, librarians, teachers and, often in the US, school boards and parent groups. These layers of oversight are the opposite of what students encounter on an Internet where anyone can post anything without establishing credentials.

Moreover, school teaches students to approach sources from beginning to end, top to bottom, before rendering judgment. Anything less is seen as rash. This intervention delivered a different message: The way to judge the credibility of a digital text is *not* by reading every word but by leaving it after a brief scan and turning to the open Internet. This message strikes some students not only as paradoxical but as an affront to common sense (Mertens et al., 2021). Against this backdrop, the main strategy students tried to master—lateral reading—constituted a break from prior instruction and represented a different epistemology of text. Skilled online searchers understand that a text's essence is not to be found in its sentences and paragraphs but in how it connects to other texts in a vast, electronically-linked network. Understanding that the

Web is a web, and the way to use it effectively is to leverage its web-like properties (Caulfield, 2017), requires a shift in how students think about decoding the written word.

Today's high school students have at their disposal an array of staggeringly powerful digital tools, all of which can be accessed by a device that fits snugly in their back pocket. However, these mind-boggling devices are operated by young people whose minds are easily boggled. How, in a digital society, do students become informed without becoming overwhelmed?

This question was at the heart of the intervention. Government and civics classes are supposed to prepare students to assume the duties of informed citizenship (Educating for American Democracy, 2020). In thinking how this might be achieved, the curricular approach pursued here cuts a different path from the core subject matter courses that make up the last two years of the high school curriculum. The goal in such courses is the acquisition of disciplinary knowledge to prepare students for further academic study (Bruner, 1960; Schwab, 1978; Shulman, 1986; Wineburg, 2001; Young, 2013). Civic Online Reasoning, on the other hand, recognizes that the informed citizen is a generalist with gaps in background knowledge who must make "good enough" decisions under real world constraints.

Confronted with completing claims about the efficacy of facemasks, it would be wonderful if everyone had the time and expertise to compute the Brownian formula of particle motion (Taylor, 1975) in order to understand how the weave of an N95 mask, with an aperture of 0.3 microns, prevents the entry of particles much tinier. But for average citizens, the decision to don a face mask is just one of innumerable decisions they must make—decisions squeezed into the brief windows between shopping for groceries, doing the laundry, preparing dinner, and getting ready for the next workday. The layperson's time is better spent evaluating the reputation of the individual or organization making a claim rather than tracking down original journal articles and reanalyzing their data (Hertwig & Engel, 2016; Jack, 2014; Origgi, 2012; Pitney, 2015)—especially when their knowledge of statistics rarely goes beyond the difference between the mean and the median.

Ecological rationality (Gigerenzer, 2001; Gigerenzer & Todd, 2012) recognizes the many constraints of everyday life. It helps us cope with a surfeit of information by identifying heuristics that simplify decision-making. It allows us to ignore unworthy information while maintaining a commitment to accuracy. In ecologically rational terms, information plays a functional role. We need no more and no less than what allows us to make better decisions—better, not best.

Although the heuristics students learned were based on observations with fact checkers, our goal was never to turn them into professionals. When the stakes are high, fact checkers at esteemed news outlets dig deeply to establish the probity of a claim, using advanced techniques to determine whether a message is legitimate (Silverman, 2020). At the high school level, we set a lower bar. Students solved tasks that required a range of quick judgments about different kinds of online content. We emphasized a few tractable and easily remembered heuristics. Given the limited time allotted for our intervention, the goal was never to have students achieve errorless searching. Rather, we hoped that like a driver doing a quick head check before switching lanes (Caulfield, 2017), students, by mastering a few digital habits, would avoid the most serious digital accidents. Obviously, they would also benefit by mastering other skills: e.g., conducting reverse image searches, checking archived versions of webpages, learning how to set the parameters on Google's filters, and so on. But we needed to start somewhere. Unlike mastering a sequence of Boolean operators or learning to decode a video's metadata, the basics of lateral reading draw on things students already know how to do: enter keywords and open new tabs in their browsers.

Skills and knowledge are inseparable. No matter how many heuristics students might apply, vetting the quality of digital information will always load on background knowledge. Competent searchers bring to search at least three kinds of background knowledge: first, knowledge specific to the Internet (how search engine optimization affects the order of results, how to mine brief snippets, the problems with relying on flawed signals of credibility, etc.); second, familiarity with trustworthy fact-checking sites like Snopes and PolitiFact, as well as the ability to discern reputable news outlets from influence peddlers, even when they appear identical (Allcott & Gentzkow, 2017); and third, the factual and conceptual knowledge that schools have been teaching since time immemorial. Lacking requisite background knowledge, students who engage in lateral reading can still get lost "because they lack domain knowledge to make sense of the information provided" (Lurie & Mustafaraj, 2018, p. 113).

Beyond this, students need to understand that the Web plays by different rules from the "friendly" texts provided to them in school. The listed author of a digital text may not be its author; an organization claiming to provide nonpartisan data might be backed by a political lobby; scientific references appended to the bottom of an article may have little bearing on the article's claims; links to authoritative sources may actually make the opposite point of what the linking site claims.

Epistemic humility (Dormandy, 2018; Hardwig, 1985) results from a sober acquaintance with one's vulnerability. No one—not even academics holding advanced degrees—is immune to the slippery ruses plied by today's digital rogues (Steinmetz, 2018). In contrast to thinking that we are smart enough to outsmart the Web, that we possess the tools to suss out a cloaked site by dissecting its About page or by detecting logical flaws in its arguments, the act of leaving an unfamiliar site to leverage the power of the open Web wrests control from a site's designer and puts it back in our own hands. We are all the wiser when we draw on the wisdom of the network rather than trying to brave it alone.

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